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ResearchGate: researchgate.net/profile/Giuseppe-Orlando-6

# Giuseppe Orlando

*Curriculum Vitae*

## Personal information

Born in GAGLIANO DEL CAPO (LE), ITALY, 18/10/1994
Citizenship: ITALIAN

## Education - Work experience

- 01/10/2023 - Present: **Postdoctoral researcher**
  *Institution*: École polytechnique (Palaiseau, France)
  *Laboratory and group*: CMAP (Centre de Mathématiques Appliquées), HPC@Maths group
  *Funding*: OPEN NUM DEF project (PI Prof. Marc Massot, Dr. Loïc Gouarin, Dr. Teddy Pichard (Maître de Conférences))
  *Supervisor*: Prof. Marc Massot
  *Research topic*: Development of numerical methods for liquid-gas two-phase flows with multi-scale modelling.
- 01/02/2023 - 30/09/2023: **Research fellowship (post-doc)**
  *Institution*: Politecnico di Milano (Milan, Italy)
  *Laboratory*: MOX - Department of Mathematics
  *Funding*: ESCAPE-2 H2020, projet Grant Agreement No. 800897
  *Supervisor*: Prof. Luca Bonaventura
  *Research topic*: Development of discontinuous finite elements codes for atmosphere dynamics simulations in the framework of `deal.II` library.
- 01/11/2019 - 31/01/2023: **Phd candidate** for the programme MATHEMATICAL MODELS AND METHODS IN ENGINEERING
  *Institution*: Politecnico di Milano (Milan, Italy)
  *Laboratory*: MOX - Department of Mathematics
  *Supervisor*: Prof. Luca Bonaventura
  *Co-supervisor*: Prof. Paolo Francesco Barbante
  *Research project*: Modelling and simulations of two-phase flows including geometric variables
  *Defence*: 18/04/2023, Milan, Italy
  *Commission*:

| | | |
|---|---|---|
| Prof. Davide Carlo Ambrosi | Politecnico di Torino | President |
| Prof. Simone Scacchi | Università degli Studi di Milano | Member |
| Prof. Luca Formaggia | Politecnico di Milano | Member |

- 29/11/2021 - 15/07/2022: **Visiting student at CMAP** (Centre de Mathématiques Appliquées), École polytechnique (Palaiseau, France)
- 2016 - 2019: **Master of Science Degree** in MATHEMATICAL ENGINEERING
  *Track*: Computational Science and Engineering
  *Institution*: Politecnico di Milano (Milan, Italy)
  *Thesis*: Development of a reactive numerical solver based on the SU2 CFD code (discussion 03/10/2019)
  *Supervisor*: Prof. Paolo Francesco Barbante
  *Mark*: 110/110
- 2013 - 2016: **Bachelor of Science Degree** in MATHEMATICAL ENGINEERING
  *Institution*: Politecnico di Milano (Milan, Italy)
  *Thesis*: Equazioni ellittiche degeneri strutturate su campi di Hörmander (discussion 22/09/2016)
  *Supervisor*: Prof. Marco Bramanti
  *Mark*: 107/110

- 2013: **Diploma**: General High School
  *Institution*: Istruzione di Istituto Superiore G. Stampacchia (Tricase (LE), Italy)
  *Mark*: 100L/100

## Training Programmes

- 2018: Introductory course on **GPU** computing for Fluid Dynamics applications
  `https://github.com/giuseppeorlando/MilanoGPU2018`
- 2019: Crash course on Vehicle Aerodynamics

## Technical skills

- Programming languages
  - **C (MPI), C++, MATLAB, FreeFem++, Python**: Academic level with the development of small projects
    `https://github.com/giuseppeorlando/Progetto_ANEDP`
    `https://github.com/giuseppeorlando/TwoPhase_Flow`
    C++ code for incompressible Navier-Stokes equations included in the official `deal.II` code-gallery
    (`https://dealii.org/developer/doxygen/deal.II/code_gallery_NavierStokes_TRBDF2_DG.html`)
    Development of codes in the open-source framework `samurai`
    (`https://github.com/hpc-maths/2025_09_two_fluid_all_topology`)
  - R, Mathematica: Basic level
- Other
  **Jupyter**, **Paraview**: Academic level

## Language skills

- Italian: Mother language
- British English: B2 Level, TOEFL iBT Certificate, Mark 86/120
- French: Written: fair, Spoken: fair

## Computer skills

- Operating systems
  Windows, **Linux**
- Text and spreadsheets
  Microsoft Office, **LaTeX**

## Research interests

- Main research interests: Computational Fluid Dynamics (**CFD**), Discontinuous Galerkin (**DG**) methods, Two-phase flows, low Mach flows, **IMEX** time discretization methods, **variational** modelling
- Other research interests: Monotonization techniques, Numerical Weather Predictions (**NWP**), shallow water equations (**SWE**), lava flow models, Scientific Machine Learning (**SciML**)

## Outline of research activity

My research activity has been mainly focused on the field of Computational Fluid Dynamics (**CFD**), spanning a wide range of applications, starting from my Master thesis on the development of a numerical solver for reactive flows, with a particular emphasis on approaches which are robust in the case of vanishing species.
During my PhD studies [21], I started to focus on the modelling and simulations of **two-phase flows**, which represents my actual main research topic. In particular, I took into account geometric variables, such as the interfacial area density, whose accurate computation is a prerequisite in order to obtain reliable values of phase exchange terms [7], and the interface mean curvature [16]. This allowed me to spend some months abroad and to start a collaboration with Centre de Mathématiques Appliquées (CMAP) at École polytechnique, where I am actually a postdoctoral researcher.
My first publication [12] deals with the development of an efficient numerical scheme for the incompressible Navier-Stokes equations. The proposed approach combines flexible discontinuous finite element space discretizations with efficient semi-implicit time discretization methods and employs an artificial compressibility formulation in order to overcome some well known issues of projection methods. More recently, an extension of this approach to two-phase flows has been proposed [6].
The first significant contribution to the literature is presented in [11] and represents a non straightforward extension to the compressible Navier-Stokes equations of real gases of **IMEX** approaches originally employed for ideal gases. More in detail, a suitable fixed point loop has been introduced for the general cubic equation of state, which can effectively handle experimental data. A discontinuous Galerkin (**DG**) spatial discretization has been employed, so as to easily achieve a full second-order method. Moreover, suitable refinement indicators have been introduced for non-ideal gases, so as to effectively use Adaptive Mesh Refinement (**AMR**). An analysis of the employed second order IMEX-RK method has been carried out, providing an improvement in terms of absolute monotonicity.
I also focused on the application of the IMEX scheme proposed in [11] to atmosphere dynamics and Numerical Weather Prediction (**NWP**) [9], showing the effectiveness of the method in a number of test cases even with non-trivial and real orography profiles. The efficiency of the proposed method and solver have been demonstrated in [14, 15]. Moreover, for the first time in the literature, the robustness of the method using non-conforming meshes [5] and curved elements [4] for flows over orography has been proven. Finally, an analysis of a general class of IMEX-RK methods in terms of asymptotic-preserving (**AP**) property has been recently performed [3] and higher order methods with respect to that employed in [11] have been validated [2, 3].

In addition, I developed a filtering monotonization technique [10] in order to avoid (or at least reduce) the oscillations that arise when high order discretization methods are used. Its potentialities have been shown in a number of benchmarks, including test cases for the Euler equations of gas dynamics.

Furthermore, I also tested for the first time in the literature to use **language models** for the development of solvers for PDEs based on the finite element method [8].

Nowadays, I am mainly working on the modelling and simulations of **two-phase flows**, with particular emphasis on the derivation of models through **variational** principles [1, 13, 19] and on the analysis of a hierarchy of Baer-Nunziato type models for compressible two-phase flows [17]. Furthermore, I am working on the development of robust and accurate models for the shallow water (de Saint-Venant) equations [20], including lava flow models [18].

## Scientific publications

### List of publications


[1] W. Haegeman, G. Orlando, S. Kokh, and M. Massot. "An all-topology two-fluid model for two-phase flows derived through Hamilton's Stationary Action Principle". *Proceedings of the Royal Society A* 482 (2026), p. 20250835. DOI: 10.1098/rspa.2025.0835.

[2] G. Orlando, S. Boscarino, and G. Russo. "A quantitative comparison of high-order asymptotic-preserving

and asymptotically-accurate IMEX methods for the Euler equations with non-ideal gases". *Computer Methods in Applied Mechanics and Engineering* 442 (2025), p. 118037. DOI: 10.1016/j.cma.2025.118037.

[3] G. Orlando and L. Bonaventura. "Asymptotic-preserving IMEX schemes for the Euler equations of non-ideal gases". *Journal of Computational Physics* 529 (2025), p. 113889. DOI: 10.1016/j.jcp.2025.113889.

[4] G. Orlando, T. Benacchio, and L. Bonaventura. "Impact of curved elements for flows over orography with a Discontinuous Galerkin scheme". *Journal of Computational Physics* 519 (2024), p. 113445. DOI: 10.1016/j.jcp.2024.113445.

[5] G. Orlando, T. Benacchio, and L. Bonaventura. "Robust and accurate simulations of flows over orography using non-conforming meshes". *Quarterly Journal of the Royal Meteorological Society* 150 (2024), pp. 4750–4770. DOI: 10.1002/qj.4839.

[6] G. Orlando. "An implicit DG solver for incompressible two-phase flows with an artificial compressibility formulation". *International Journal for Numerical Methods in Fluids* 96 (2024), pp. 1932–1959. DOI: 10.1002/fld.5328. Preprint: arXiv: 2307.04580.

[7] G. Orlando, P.F. Barbante, and L. Bonaventura. "On the evolution equations of interfacial variables in two-phase flows". *International Journal of Multiphase Flow* 169 (2023), p. 104617. DOI: 10.1016/j.ijmultiphaseflow.2023.104617.

[8] G. Orlando. "Assessing ChatGPT for coding finite element methods". *Journal of Machine Learning for Modeling and Computing* (2023). DOI: 10.1615/JMachLearnModelComput.2023049326.

[9] G. Orlando, T. Benacchio, and L. Bonaventura. "An IMEX-DG solver for atmospheric dynamics simulations with adaptive mesh refinement". *Journal of Computational and Applied Mathematics* (2023), p. 115124. DOI: 10.1016/j.cam.2023.115124.

[10] G. Orlando. "A filtering monotonization approach for DG discretizations of hyperbolic problems". *Computers & Mathematics with Applications* 129 (2023), pp. 113–125. DOI: 10.1016/j.camwa.2022.11.017. Preprint: arXiv: 2204.08693.

[11] G. Orlando, P.F. Barbante, and L. Bonaventura. "An efficient IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases". *Journal of Computational Physics* 471 (2022), p. 111653. DOI: 10.1016/j.jcp.2022.111653.

[12] G. Orlando et al. "An efficient and accurate implicit DG solver for the incompressible Navier-Stokes equations". *International Journal for Numerical Methods in Fluids* 94 (2022), pp. 1484–1516. DOI: 10.1002/fld.5098.

## List of proceedings

[13] W. Haegeman, S. Kokh, M. Massot, and G. Orlando. "A generic framework to derive systems of conservation laws with source terms and its application to heat conduction in fluid flows: An alternative to the method of moments in kinetic theory of gases?" *ESAIM: Proceedings and Surveys* 78 (2025), pp. 80–97. DOI: 10.1051/proc/202578080.

[14] G. Orlando, T. Benacchio, and L. Bonaventura. "Improving the scalability of a high-order atmospheric dynamics solver based on the deal.II library". *Procedia Computer Science* 267 (2025), pp. 227–236. DOI: 10.1016/j.procs.2025.08.249. Proceedings of the Third EuroHPC Day.

[15] G. Orlando, T. Benacchio, and L. Bonaventura. "Efficient and scalable atmospheric dynamics simulations using non-conforming meshes". *Procedia Computer Science* 255 (2025), pp. 33–42. DOI: 10.1016/j.procs.2025.02.258. Proceedings of the Second EuroHPC Day.

[16] G. Orlando, P. F. Barbante, and L. Bonaventura. "Assessment of Different Strategies to Compute Surface Tension Contribution". *Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 2: European Conference, September 4-8, Lisbon, Portugal*. Vol. 154. Springer Nature. 2025, pp. 213–222. DOI: 10.1007/978-3-031-86169-7_22. Preprint: arXiv: 2402.04670.

## List of preprints


[17] G. Orlando, W. Haegeman, M. Pelanti, and M. Massot. *A robust computational framework for the mixture-energy-consistent six-equation two-phase model with instantaneous mechanical relaxation terms*. 2025. arXiv: 2509.26284 [math.NA].

[18] F. Gatti and G. Orlando. *Second-order Optimally Stable IMEX (pseudo-)staggered Galerkin discretization: application to lava flow modeling*. 2025. arXiv: 2509.09460 [math.NA].

[19] W. Haegeman, G. Orlando, S. Kokh, and M. Massot. *Modelling of relative velocity, velocity fluctuations and their interactions for two-fluid models by Stationary Action Principle*. 2025. URL: \href{https://hal.science/hal-05224712}{hal-05224712}.

[20] L. Arpaia, G. Orlando, C. Ferrarin, and L. Bonaventura. *A high-order matrix-free adaptive solver for the shallow water equations with irregular bathymetry*. 2025. arXiv: 2505.18743 [cs.CE].


## PhD thesis


[21] G. Orlando. "Modelling and simulations of two-phase flows including geometric variables". http://hdl.handle.net/10589/198599. PhD thesis. Politecnico di Milano, 2023.


## Master thesis


[22] G. Orlando. *Development of a reactive numerical solver based on the SU2 CFD code*. http://hdl.handle.net/10589/150021. Master thesis, Politecnico di Milano. 2019.


## Editorial activity

- 01/01/2026 - :Associate editor for *Monthly Weather Review* (*MWR*)
- I have acted as a referee for international journals in Mathematical Modelling and Scientific Computing: *Computers and Mathematics with Applications* (*CAMWA*), *Journal of Computational Physics* (*JCP*), *Computer & Fluids* (*CAF*), *Monthly Weather Review* (*MWR*), *International Journal on Geomathematics* (*GEM*).

## Main conferences partecipation

- **Invited seminar** for the series "Séminaire Analyse Appliquée (AA)" at I2M (Institut de Mathématiques de Marseille)
  *Date and Location*: 31/03/2026, Marseille (France)
  *Title and Duration*: "An all-topology two-fluid model for two-phase flows derived through Hamilton's Stationary Action Principle", 45 minutes
- Speaker Flash workshop
  *Date and Location*: 02/02/2026, Palaiseau (France)
  *Title and Duration*: "A unified compressible two-phase, two-scale model with capillarity for separated-to-disperse transitions in atomizing flows", 50 minutes
- **Invited speaker** at 38ème séminaire de mécanique des fluides numérique
  *Date and Location*: 26/01/2026, Paris (France)
  *Title and Duration*: "A unified compressible two-phase, two-scale model with capillarity for separated-to-disperse transitions in atomizing flows", 40 minutes
- **Invited seminar** for the series "Séminaires de Mathématiques et de leurs Applications" at LMAP (Laboratoire de Mathématiques et de leurs Applications, Pau)
  *Date and Location*: 22/01/2026, Pau (France)
  *Title and Duration*: "An adaptive high-order DG solver for the shallow water equations with irregular bathymetry", 45 minutes
- **Invited speaker** at ARIA (Accurate ROMs for Industrial Applications) 2025 workshop
  *Date and Location*: 23/09/2025, Biarritz (France)
  *Title and Duration*: "A high-order matrix-free adaptive DG solver for the shallow water equations with irregular bathymetry", 30 minutes
- **Invited talk** in the minisymposium "High-Order and Innovative Methods for Coupled Problems in Life Science and Geophysics" at 11th International Conference on Coupled Problems in Science and Engineering (COUPLED 2025)

*Organizers*: F. Bonizzoni, M. Corti, and S. Pagani *Date and Location*: 27/05/2025, Villasimius (Italy)
*Title and Duration*: "A high-order solver for NWP using non-conforming adaptive andcurved meshes: applications to flows over orography", 15 minutes

- 12th International Conference on Multiphase Flow (ICMF 2025)
  *Date and Location*: 14/05/2025, Toulouse (France)
  *Title and Duration*: "A unified two-scale gas-liquid two-fluid model with capillarity and interface regularization through a mass transfer between scales", 15 minutes

- **Invited seminar** for the series "Séminaire de Calcul Scientifique et Modélisation" at IMB (Institut de Mathématiques de Bordeaux)
  *Date and Location*: 10/04/2025, Bordeaux (France)
  *Title and Duration*: "Asymptotic-preserving IMEX schemes for the Euler equations of non-ideal gases", 45 minutes

- **Invited speaker** at Seminar Mathematics and Atmospheric Physics
  *Date*: 23/10/2024
  *Title and Duration*: "Robust and accurate simulations of flows over orography using non-conforming variable resolution meshes", 45 minutes

- Seminar HPC@Maths
  *Date and Location*: 18/10/2024, Palaiseau (France)
  *Title*: "Numerical analysis of a two-fluid model"

- European Meteorological Society (EMS) Annual Meeting
  *Date and Location*: 06/09/2024, Barcelona (Spain)
  *Title and Duration*: "Robust and accurate simulations of flows over orography using non-conforming variable resolution meshes", 15 minutes

- 2nd BICTAM-CISM Symposium on Dispersed Multiphase Flows: from Measuring to Modeling
  *Date and Location*: 22/08/2024, Beijing (China)
  *Title and Duration*: "A unified two-scale gas-liquid two-fluid model with capillarity and interface regularization through a mass transfer between scales", 15 minutes

- International Conference on Numerical Methods in Multiphase Flows - 5 (ICNMMF-5)
  *Date and Location*: 28/06/2024, Reykjavik (Iceland)
  *Title and Duration*: "A two-scale model for two-phase flows including mass transfer", 15 minutes

- **Invited talk** in the minisymposium "Computational Methods for Multiphase Flows with Liquid-Vapor Transition" at 9th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS 2024)
  *Organizers*: L. Brandt, M. Pelanti, M. Rodriguez, and S. Tanguy
  *Date and Location*: 06/06/2024, Lisbon (Portugal)
  *Title and Duration*: "A two-scale model for two-phase flows including geometric variables and mass transfer", 15 minutes

- 46ème Congrès National d'Analyse Numérique (CANUM 2024)
  *Date and Location*: 29/05/2024, Ile de Ré (France)
  *Title and Duration*: "A two-scale model for two-phase flows including geometric variables and mass transfer"; 15 minutes

- **Invited speaker** at the INdAM Workshp INSiDEs (Innovations in the Numerical Treatment of Stiff Differential Equations)
  *Date and Location* 22/02/2024, Rome (Italy)
  *Title and Duration*: "An asymptotic-preserving scheme for Euler equations with non-ideal gases", 45 minutes

- Seminar HPC@Maths
  *Date and Location*: 24/11/2023, Palaiseau (France)
  *Title*: "Derivation of relations for interfacial area density in two-phase flows through the Stationary Action Principle"

- **Invited talk** in the minisymposium "Theoretical and computational aspects of discontinuous Galerkin methods" at European Conference on Numerical Mathematics and Advanced Applications (ENUMATH 2023)

*Organizers*: V. Aizinger and A. Rupp
*Date and Location*: 05/09/2023, Lisbon (Portugal)
*Title and Duration*: "A filtering monotonization technique for DG discretization of hyperbolic problems", 25 minutes

- Bi-annual congress of SIMAI 2023
  *Date and Location*: 29/08/2023, Matera (Italy)
  *Title and Duration*: "A filtering monotonization technique for DG discretization of hyperbolic problems", 25 minutes

- **Invited talk** in the minisymposium "Computational Advances in Earth System Modeling" at 17th U. S. National Congress on Computational Mechanics (USNCCM17)
  *Organizers*: M. Perego, A. Salinger, and I. Tezaur
  *Date and Location*: 25/07/2023, Albuquerque (USA)
  *Title and Duration*: "An IMEX-DG solver for atmospheric dynamics with Adaptive Mesh Refinement", 20 minutes

- PDEs on the sphere 2023 (PDEs 2023)
  *Date and Location*: 06/07/2023, Grenoble (France)
  *Title and duration*: "An IMEX-DG solver for atmospheric dynamics based on deal.II framework", 30 minutes

- 7th ECCOMAS Young Investigators Conference (YIC2023)
  *Date and Location*: 21/06/2023, Porto (Portugal)
  *Title and Duration*: "An IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases", 15 minutes

- **Invited seminar** for the series "Seminario di Modellistica Differenziale Numerica" at Sapienza Univeristá di Roma
  *Date and Location*: 22/11/2022, Rome (Italy)
  *Title and Duration*: "A filtering monotonization technique for DG discretizations of hyperbolic problems", 45 minutes

- 14th European Fluid Mechanics Conference (EFMC14)
  *Date and Location*: 14/09/2022, Athens (Greece)
  *Title and Duration*: "An IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases", 12 minutes

- XVIII International Conference on Hyperbolic Problems: Theory, Numerics, Applications (HYP2022)
  *Date and Location*: 21/06/2022, Malaga (Spain)
  *Title and Duration*: "An IMEX-DG solver for the compressible Navier-Stokes equations with a general equation of state", 15 minutes

- 45ème Congrès National d'Analyse Numérique (CANUM 2020)
  *Date and Location*: 16/06/2022, Evian-les-Bains (France)
  *Title and Duration*: "An IMEX-DG solver for the compressible Navier-Stokes equations with a general equation of state", 15 minutes

- Bi-annual congress of SIMAI 2020+2021
  *Date and Location*: 30/08/2021, Parma (Italy)
  *Title and Duration*: "An accurate and efficient implicit Discontinuous Galerkin solver for the incompressible Navier-Stokes equations", 25 minutes

- Conference "Numerical methods for hyperbolic problems (NumHyp 2021)"
  *Date*: 26/07/2021
  *Title and Duration*: "An efficient and accurate implicit DG solver for the incompressible Navier-Stokes equations", 30 minutes

## Outline of teaching activity

**Responsible teacher**

- Crash course on the use of MATLAB software for students - Spring Semester - A.Y. 2019-2020 and 2020-2021
  *Teaching load*: 16 hours (per course)
  *Institution*: Politecnico di Milano - Lecco Campus (Italy)

**Exercise Session**

- APM51656EP "Elements of measure theory and functional analysis" - A.Y. 2025-2026
  *Teaching load*: 6 hours
  *Institution*: École polytechnique (France)
  *Path*: Master Applied Mathematics and Statistics (M1)
  *Responsible of the course*: Prof. David Gérard-Varet

- MECCANICA RAZIONALE - A.Y. 2020-2021
  *Teaching load*: 20 hours
  *Institution*: Politecnico di Milano (Italy)
  *Path*: Bachelor degree in Mathematical Engineering (Italian)
  *Responsible of the course*: Prof. Maurizio Stefano Vianello

- MATHEMATICAL MODELS AND METHODS IN ENGINEERING [1] - A.Y. 2019-2020 and 2020-2021
  *Teaching load*: 10 hours (A.Y. 2019-2020), 20 hours (A.Y. 2020-2021)
  *Institution*: Politecnico di Milano (Italy)
  *Path*: Master degree in Mathematical Engineering
  *Responsible of the course*: Prof. Maurizio Stefano Vianello

**Informatic Laboratory**

- **Informatic Laboratory** for the course SHAPES AND ALGORITHMS: TOOLS FOR GENERATIVE DESIGN - A.Y. 2020-2021
  *Teaching load*: 26 hours
  *Institution*: Politecnico di Milano (Italy) *Responsible of the course*: Prof. Anna Scotti

**Tutoring**

- **Tutor of the final project** for the APM51051EP course "Systémes dynamiques pour la modèlisation et la simulation des milieux rèactifs multi-èchelles" (M2) - A.Y. 2024-2025, 2025-2026
  *Institution*: École polytechnique (France)
  *Responsible of the course*: Prof. Marc Massot

- **Tutor of the final project** for the MAP551 course "Systémes dynamiques pour la modèlisation et la simulation des milieux rèactifs multi-èchelles" (M2) - A.Y. 2023-2024
  *Institution*: École polytechnique (France)
  *Responsible of the course*: Prof. Marc Massot

## Supervision activity Master thesis

1. Safira Leandra Liora Piasko (September 2022 - March 2023)
   *Institution*: ETH Zurich (Swiss)
   *Title*: Discontinuous Finite Element Methods for Large Eddy Simulation(LES) of Incompressible Flows
   *Collaborators*: Prof. Luca Bonaventura, Prof. Antonella Abbà, Prof. Christoph Schär

2. Letizia Bottani (September 2025 - February 2026)
   *Institution*: Double program Politecnico di Milano (Italy) - Technical University of Denmark (Denmark) *Title*: Simulation of rotating flows with a high-order finite element framework for atmospheric dynamics with adaptive mesh refinement
   *Collaborators*: Prof. Luca Bonaventura, Dr. Tommaso Benacchio, Prof. Allan Peter Engsig-Karup

## Projects, organizations and other info

- 17/11/2025: **Co-organiser** of of the practical training session "Hands-on Samurai", dedicated to introducing the `samurai` code to approximately 25 external participants from 10 different research institutions (held on site at CMAP)

- 07/07/2025 - 01/08/2025 : Participation to Modeling Summer Visit (MSV) Program as co-PI of a research project.
  *Title*: A two-phase flow model with scale separation for the separated-to-disperse phase transition
  *Institution*: NASA Ames
  *Funding*: Partial support by Fondation Mathématique Jacques Hadamard (2k €) through the program Junior Scientific Visibility

- **Co-organization of seminars HPC@Maths** at CMAP - École polytechnique starting from November 2023

- **Co-organisation of a minisymposium** at SIMAI 2025 Congress
  *Title*: Recent advancements in numerical methods and physical modelling of multi-material flows
  *Co-organizers*: Niccolò Tonicello (SISSA), Federico Dalla Barba (Padua University), Marica Pelanti (ENSTA)

- 28/08/2023 - 01/09/2023: **Staff member** of Bi-annual congress of SIMAI 2023

- Solver for incompressible Navier-Stokes equations included in the **official** `deal.II` **code-gallery**
  `https://dealii.org/developer/doxygen/deal.II/code_gallery_NavierStokes_TRBDF2_DG.html`

- PI of the EuroHPC Benchmark project EHPC-BEN-2024B03-045
  *Title*: A high-order efficient and scalable solver for atmospheric dynamics using non-conforming and curved meshes

- PI of the EuroHPC Development project EHPC-DEV-2024D10-054
  *Title*: A high-order efficient and scalable solver for NWP using dynamically adaptive non-conforming meshes

- PI of the ISCRA-C Project FEM-GPU
  *Title*: Finite Element Methods for GPU architectures

- PI of the ISCRA-C Project NUMNETF
  *Title*: NUmerical Methods for Non-Equilibrium Two-phase Flows
  *Research Group*: Paolo Francesco Barbante, Luca Bonaventura
  *Amount*: 1 720 €(=86k core hours on G100 cluster, CINECA, Italy)

- PI of the ISCRA-C Project SIDICONS
  *Title*: Semi-Implicit Discretizations for Incompressible and COmpressible Navier-Stokes equations
  *Research group*: Antonella Abbá, Paolo Francesco Barbante, Luca Bonaventura

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